Ahmad Rayan Qasim – Personal and Professional Profile

This profile compiles the personal, academic, and extracurricular details of Ahmad Rayan Qasim, presented in structured sections for clarity. It highlights his educational journey, achievements, projects, leadership roles, and skills.

Personal Information

- **Full Name**: Ahmad Rayan Qasim
- **Mobile Number**: +92 316 6461653
- **Email Address**: iamahmadrayan@gmail.com
- **LinkedIn**: https://www.linkedin.com/in/ahmad-rayan-2a1785261/
- **Portfolio Website**: https://ahmadrayanqasim.vercel.app/
- **Location**: Multan, Pakistan

Educational Background

O-Levels – La Salle Higher Secondary School, Multan

- Achieved strong academic performance with consistent grades.
- Actively participated in sports including badminton, football, and cricket.
- Organized and managed 10 stalls with peers during the school funfair.
- Twice awarded **Science Fair Winner**:
  - **8th Class**: Designed and built an incubator project.
  - **9th Class**: Created a thermal power station using mirrors to generate steam and electricity.
- Developed a sensor device using Arduino and buzzer technology to detect a 5-foot distance, designed to assist visually impaired individuals and promote social distancing during COVID-19.

A-Levels (2022–2024)

A1 – Bloomfield Hall, Multan Cantt

- Excelled academically; received free personal tutoring from a teacher in recognition of performance.
- Continued excellence in badminton.

A2 – Garrison Academy, Multan

- Selected as coach for the girls' badminton team at the sports gala; team won under his training.
- Remained undefeated in badminton throughout college.
- Awarded the title **"Mr. Evening"** at the farewell ceremony.
- Achieved strong grades in Mathematics, Chemistry, and Physics.
- Represented Germany as delegation head in **Model United Nations (MUN)**.
- Led a team of 11 members in the **Global Village event**, showcasing German culture.

Bachelor's Degree – Computer Science (2024–2028, Air University Multan)

- **Academic Performance:** Enrolled in BS Computer Science with a current CGPA of **3.55** (5th semester ongoing). Achieved maximum GPA (12/12 grade points) in Data Structures & Algorithms.
- **Leadership & Model UN:**
    - **Air University Model United Nations:** Served as **Academics Head** and **Chair of UNHRC**.
    - **Debating Society:** Appointed **Team Coordinator**, driving public speaking and team management.
    - **Tech Bist Society:** Selected as **Core Member** (2024–2025).
    - **Startup Club & Business Incubation Center (BIC):** Elected **President** of the Air University Startup Club (2026–2027 term) in coordination with the Business Incubation Center (BIC), where he manages and guides multiple student startups, mentors entrepreneurial ventures, and coordinates strategic startup growth.
- **Events & Entrepreneurship:** Managed a Summer Fest stall, achieving a **50% profit margin**, alongside active participation in campus events.

Student Details & Academic Transcript

- **Name:** Ahmad Rayan Qasim
- **Reg ID:** 243557
- **Overall CGPA:** 3.55
- **Current Semester (Ongoing):** 5th

Semester 1 (FALL-2024) — *SGPA: 3.51*

- Applications of Information and Communication Technologies — 2 Cr. Hrs | Grade: A | GP: 8
- Applications of Information and Communication Technologies Lab — 1 Cr. Hr | Grade: A | GP: 4
- Applied Physics — 2 Cr. Hrs | Grade: B- | GP: 5.34
- Applied Physics Lab — 1 Cr. Hr | Grade: B+ | GP: 3.33
- Functional English — 3 Cr. Hrs | Grade: B+ | GP: 9.99
- Ideology and Constitution of Pakistan — 2 Cr. Hrs | Grade: A- | GP: 7.34
- Programming Fundamentals — 3 Cr. Hrs | Grade: A- | GP: 11.01

- Programming Fundamentals Lab — 1 Cr. Hr | Grade: A- | GP: 3.67

Semester 2 (SPRING-2025) — *SGPA: 3.47*

- Database Systems — 3 Cr. Hrs | Grade: A- | GP: 11.01
- Database Systems Lab — 1 Cr. Hr | Grade: A- | GP: 3.67
- Digital Logic Design — 2 Cr. Hrs | Grade: B+ | GP: 6.66
- Digital Logic Design Lab — 1 Cr. Hr | Grade: B+ | GP: 3.33
- Discrete Structures — 3 Cr. Hrs | Grade: A- | GP: 11.01
- Expository Writing — 3 Cr. Hrs | Grade: A- | GP: 11.01
- Object Oriented Programming — 3 Cr. Hrs | Grade: B | GP: 9
- Object Oriented Programming Lab — 1 Cr. Hr | Grade: B+ | GP: 3.33
- Quranic Studies — 0 Cr. Hrs | Grade: S | GP: 0
- Sirat Ul Nabi — 0 Cr. Hrs | Grade: S | GP: 0

Semester 3 (FALL-2025) — *SGPA: 3.69*

- Calculus and Analytical Geometry — 3 Cr. Hrs | Grade: A- | GP: 11.01
- Data Structures — 3 Cr. Hrs | Grade: A | GP: 12
- Data Structures Lab — 1 Cr. Hr | Grade: A | GP: 4
- Information Security — 2 Cr. Hrs | Grade: A- | GP: 7.34
- Information Security Lab — 1 Cr. Hr | Grade: A- | GP: 3.67
- Probability and Statistics — 3 Cr. Hrs | Grade: B+ | GP: 9.99
- Quranic Studies — 0 Cr. Hrs | Grade: S | GP: 0
- Sirat Ul Nabi — 0 Cr. Hrs | Grade: S | GP: 0
- Visual Programming — 2 Cr. Hrs | Grade: A- | GP: 7.34
- Visual Programming Lab — 1 Cr. Hr | Grade: A- | GP: 3.67

Semester 4 (SPRING-2026) — *SGPA: 3.53*

- Civics and Community Engagement — 2 Cr. Hrs | Grade: A- | GP: 7.34
- Computer Networks — 2 Cr. Hrs | Grade: B+ | GP: 6.66
- Computer Networks Lab — 1 Cr. Hr | Grade: B+ | GP: 3.33
- Computer Organization and Assembly Language — 2 Cr. Hrs | Grade: A- | GP: 7.34
- Computer Organization and Assembly Language Lab — 1 Cr. Hr | Grade: A | GP: 4
- Full Stack Web Development — 2 Cr. Hrs | Grade: B+ | GP: 6.66 - MERN Stack
- Full Stack Web Development Lab — 1 Cr. Hr | Grade: A- | GP: 3.67 - MERN Stack
- Multivariable Calculus — 3 Cr. Hrs | Grade: A- | GP: 11.01
- Software Engineering — 3 Cr. Hrs | Grade: B+ | GP: 9.99

Major Wins & Competitions

- **1st Place – Hacktoberfest Hack Day Multan x AWS Student Builder Group BZU (October 31, 2026)**: Awarded 1st place in an international hackathon organized by Major League Hacking (MLH) and Devpost, bringing together teams from all over South Punjab. Built and live **MandiLinkAI** under a strict 60-minute time constraint using Gemini AI, Snowflake CoCo, Solana Agent Registry, and GitHub Copilot.

Projects – Ahmad Rayan Qasim

University Projects

1st Semester

- **Programming Fundamentals**: Developed an **Inventory Management System** using C++.

2nd Semester

Database Management Systems (CS130 & CS130L)

- **Project**: Cattle Farm Management System – Relational DBMS Automation
    - **Tech Stack**: SQL Server, Visual Studio (C# front-end), stored procedures, triggers, user-defined functions.
    - Designed a relational database with **15+ normalized tables**, enforcing referential integrity and constraints to centralize animal, user, financial, and inventory records.
    - Automated workflows via **54 stored procedures, 18 functions, and 17 triggers**, enabling real-time vet assignment, breeding cycle prediction, expense logging, and archival processes.
    - Integrated front-end forms with backend procedures, reducing manual errors and improving reporting accuracy by **20%**.
    - Delivered actionable insights through dashboards and functions (e.g., fn_CalculateMonthlyProfit, fn_IsBreedingEligible).
    - Received **highest marks in the batch**.

Object-Oriented Programming (C++)

- **Project**: Cattle Farm Management System – Object-Oriented Design & Financial Integration
    - **Tech Stack**: C++ (OOP classes, inheritance, operator overloading), file I/O, modular architecture.
    - Built a farm management simulator with modular managers for animals, tasks, users, and financial records.
    - Applied advanced OOP principles (inheritance, encapsulation, operator overloading).
    - Implemented animal lifecycle management with specialized subclasses (cow, bull, calf, meat animal).
    - Developed financial reporting modules for expense/income tracking and profit calculation.
    - Recognized as the **best OOP project in class**, personally appreciated by the instructor.

Digital Logic and Design

- Constructed a complete working computer with all components simulated in **Logisim**.
- Project received **highest marks**; instructor was highly impressed.

3rd Semester

Visual Programming

- **Project**: CodeToSurvive – Gamified Visual Programming Simulator
    - **Tech Stack**: C# (.NET 8.0), WPF, EF Core 8.0, SQL Server LocalDB.
    - Designed an immersive gamified learning platform simulating a developer's journey with coding challenges, financial decision-making, and lifestyle management.
    - Built custom syntax parser (Tokenizer & Parser classes) to validate code submissions.
    - Achieved **80% unit test coverage** with xUnit; conducted integration and UI tests.
    - Future plans: AI-driven adaptive challenges, cloud deployment, mobile port, multiplayer features.

Probability & Statistics

- **Project**: Financial Pulse Analyzer – Statistical Modeling of Student Economy
    - **Tech Stack**: R, SPSS, Minitab, Excel.
    - Designed a dataset of 19 students via questionnaire.
    - Applied statistical workflows (distributions, regression, hypothesis testing).
    - Delivered insights: negative correlation between spending and savings, moderate budgeting confidence, age-linked saving patterns.

Data Structures & Algorithms (DSA)

- **Project**: SecureShell – Enhanced Command-Line Shell

        - Built with C++ using tries, bloom filters, treaps, hash tables.
        - Features: autocompletion, malware detection, prioritized history, secure command registry.
        - Validated with unit tests (500 trie inserts/suggestions, bloom false positives <5%).
    - **Project**: Round Robin CPU Scheduler
        - Implemented in C++ with manual queues, merge sort, pointers.
        - Simulated CPU scheduling with Gantt chart output.
        - Delivered metrics: avg waiting time 11.5 ms, turnaround 18.25 ms, CPU utilization 100%.

Information Security

- **Project**: Secure Data Toolbox – Hashing, Cryptography & Steganography
    - **Tech Stack**: C# .NET (Windows Forms), System.Security.Cryptography, System.Drawing.
    - Integrated SHA-256 hashing, PBKDF2 salting, classical ciphers, and LSB steganography.
    - Delivered forensic-ready outputs with reproducible hash values and tamper detection.

4th Semester

Major Project: LawLedge (Legal Portal & Community Grievance System)

- **Type:** Full-Stack Web Application (SPA + Real-Time WebSocket Server)
- **Team Members:** Ahmad Rayan Qasim (243557), Abeera Shahid, Attique Alvi, Huda Saeed
- **Supervisor:** Prof. Muhammad Rashaf Jamil Khan

Core Problem Solved:

- Addressed the lack of transparency, structural friction, and slow communication in traditional citizen grievance reporting by building an end-to-end digital portal that automates case routing, evidence management, and real-time community collaboration.

Technical Stack & Architecture:

- **Frontend:** React 18, Vite, Tailwind CSS, Framer Motion (animated module shell & route transitions), React Router DOM (role-based protected routing), Lucide React.
- **Backend & Real-Time Server:** Node.js, Express, Socket.IO (persistent WebSocket hub for live messaging, online status broadcasts, and event triggers).
- **Database & Cloud Storage:** Supabase (PostgreSQL relational database, secure Auth session management, and encrypted file storage buckets for evidence uploads).
- **Client-Side Utilities:** jsPDF & jspdf-autotable (automated generation of official, authority-templated PDF complaint reports with embedded evidence images), EmailJS.

Key Engineering & Architectural Highlights:

- **Role-Based Access Control (RBAC) & Security:** Implemented robust session handling via Supabase Auth with multi-layered role resolution (User, Volunteer, Admin) and instant admin account-blocking logic to secure restricted endpoints.
- **Real-Time WebSocket Communication:** Architected a custom Socket.IO server (lawledge-server) supporting direct messaging, group rooms, and real-time broadcasts for emergency help requests and community posts.
- **Automated PDF Generation Pipeline:** Built a client-side document engine that compiles form inputs, dynamic authority color schemes, and cloud-hosted evidence URLs into downloadable official complaint filings without server-side rendering latency.
- **Production Deployment Configuration:** Structured a clean separation of concerns between a Vite-optimized SPA frontend (deployed on Vercel via vercel.json) and a dedicated Node/Socket.IO backend instance.

CodeConnect — Bridging High-Level UI with Low-Level x86 Logic

- **Context & Team:** CS226: Computer Org & Assembly | Air University | Instructor: Ms. Mamoona Rafique Khan | Team: Ahmad Rayan Qasim (Lead Integrator), Attique Alvi, Faisal Imam, Shahzaib Riaz
- **Overview:** Gamified puzzle platform transforming x86 assembly instructions into connectable visual nodes to teach register management and memory operations.
- **Tech Stack:** C# WinForms (CodeConnectUI), P/Invoke DllImport (__cdecl/__stdcall), 32-bit x86 Assembly DLL (backend.asm).
- **Key Highlights:** Double-buffered WinForms UI; 300-byte data segment (times 300 db 0) for a 20x15 grid via base-index addressing ([grid+eax]); 4-bit bitwise packing for FlowID/NodeType; strict x86 stack frames (push ebp) & register preservation (pusha/popa).
- **Problem-Solving:** Fixed BadImageFormatException by forcing x86 Visual Studio build target; allocated strict StringBuilder capacity to eliminate C#/ASM string memory segfaults.
- **Impact:** Fused managed desktop frameworks with unmanaged low-level x86 logic for interactive systems education.

Transportation and Logistics Network Optimization

- **Context & Team:** Computer Networks Project | Air University | Instructor: Sir Mughees | Team: Ahmad Rayan Qasim, M. Attique Alvi, Faisal Imam, Shahzaib Riaz
- **Overview:** Multi-site enterprise network architecture isolating operational subnets from public access while optimizing inter-hub traffic and bandwidth allocation.
- **Tech Stack:** Cisco 2811 Routers, 2960 Switches, ASA 5506-X Firewalls, WAPs, 802.1Q, Extended ACLs, CBWFQ, LLQ, Cisco IOS CLI, FTP.
- **Key Highlights:** Logical VLAN segmentation (VLAN 10 Trans: 192.168.10.0/24, VLAN 20 Logistics: 192.168.20.0/24, VLAN 30 Guest: 192.168.30.0/24) via 802.1Q trunking; Extended ACLs (100, 110); CBWFQ/LLQ allocating 50% priority bandwidth to Ops and capping Guest at 20%; Central FTP/DB server (192.168.10.15).
- **Problem-Solving:** Deployed outbound ACLs to drop unauthorized guest access attempts; applied dynamic priority queues to fix real-time logistics latency bottlenecks.

- **Impact:** Enterprise-grade model demonstrating advanced Cisco routing, switching, QoS, and security configuration.

VectorGlide 3D — Real-Time Parametric Trajectory Simulation Engine

- **Context & Team:** Multivariable Calculus Project | Air University Multan | Guided by Dr. Ammara Khakhwani | Team: Ahmad Rayan Qasim, M. Attique Alvi, Abdul Kabir Syed, Abdul Samad
- **Overview:** 3D web simulation platform modeling autonomous drone trajectories via space curve calculus to visualize kinematic paths and MPC navigation telemetry.
- **Tech Stack:** React, Three.js, React Three Fiber (DroneScene.jsx), Math.js (CalculusEngine.js), Blender (Drome_Motion_Model.blend), GLTF.
- **Key Highlights:** Evaluates $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, velocity $r'(t)$, speed $\vert{}\mathbf{r}'(t)\vert{}$, and arc length s; renders 500+ point curves with Frenet-Serret frames ($T(t)$, $N(t)$) and coordinate plane projections; arc-length reparameterization for constant speed; custom Blender low-poly GLTF drone asset.
- **Problem-Solving:** Calculated frame rotations along $T(t)/N(t)$ to eliminate turn singularities; built a 1080p WebGL canvas + telemetry snapshot exporter.
- **Impact:** Practical application of differential geometry for drone path planning, structural stress forecasting, and battery optimization.

ALL Github Projects Repos

Imp: username = “ahmadrayan-create”

Production Usage Metering & Billing Engine

- **1-Line Summary:** An enterprise usage-based billing platform engineered for SaaS providers to accurately meter API consumption, enforce quotas, and automate payment processing.
- **Tech Stack:** Python, FastAPI, PostgreSQL, Stripe API, Docker, Pydantic
- **Key Features / What I Built:**
    - Engineered an asynchronous HTTP metering engine with integer-based micro-cent precision to eliminate floating-point financial rounding errors.
    - Implemented strict idempotency keys and signature-verification middleware to secure Stripe webhook ingestion pipelines.
    - Built a dynamic quota-enforcement engine that evaluates customer API usage tiers in real time prior to route execution.
    - Structured ACID-compliant PostgreSQL transactions to record high-throughput transactional event logs.
- **Problem Solved / Impact:**
    - Prevented revenue leakage and double-billing vulnerabilities by enforcing exactly-once processing across micro-cent API transactions.

    - Maintained low-latency response times during peak concurrent traffic while ensuring an audit-grade billing history.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/flyrank-capstone-metering-billing](https://github.com/username/flyrank-capstone-metering-billing)

Automated LLM Support Triage API

- **1-Line Summary:** An intelligent backend classification microservice built for enterprise customer support teams to automatically triage, validate, and route incoming tickets using LLMs.
- **Tech Stack:** Python, FastAPI, OpenRouter API, Pydantic, Asyncio
- **Key Features / What I Built:**
    - Architected asynchronous REST endpoints leveraging OpenRouter LLM integrations for real-time query intent parsing.
    - Implemented strict Pydantic schema validation coupled with automated self-healing retry loops for malformed model outputs.
    - Developed a quarantine logging mechanism to isolate low-confidence inferences and non-deterministic schema failures.
    - Configured structured JSON response formatting to guarantee seamless downstream API consumption.
- **Problem Solved / Impact:**
    - Reduced ticket triage latency from hours to milliseconds while maintaining a zero-error rate for schema validation.
    - Lowered API call retry overhead by 35% through dynamic local payload repair loops before triggering fallback routes.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/llm-triage-api](https://github.com/username/llm-triage-api)

Enterprise Electronic Logging Device (ELD) & Compliance Engine

- **1-Line Summary:** A full-stack commercial fleet navigation and compliance engine built for logistics operators to automate Hours of Service (HOS) routing and FMCSA reporting.
- **Tech Stack:** TypeScript, React, Django, Python, Leaflet.js, SVG, Tailwind CSS
- **Key Features / What I Built:**
    - Designed an interactive GIS routing dashboard using Leaflet to calculate commercial transit paths and fuel stops across long-haul routes.
    - Built an automated compliance calculation engine adhering strictly to FMCSA 11-hour driving and 14-hour duty regulations.
    - Generated dynamic vector SVG 24-hour log sheets providing real-time visual representation of driver duty-status changes.
    - Integrated automated rest-break triggers and loading-stop schedulers directly into route itinerary pipelines.
- **Problem Solved / Impact:**
    - Replaced manual paper-log tracking, eliminating non-compliance penalties for long-haul commercial motor carriers.

    - Streamlined route planning efficiency, significantly reducing driver downtime during multi-state transit schedules.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/eld-compliance-engine](https://github.com/username/eld-compliance-engine)

Resilient & Ethical Web Scraping Pipeline

- **1-Line Summary:** An enterprise-grade, ethically constrained web extraction pipeline engineered for data teams to scrape target sites safely without triggering IP blocks.
- **Tech Stack:** Python, BeautifulSoup4, Pydantic, Asyncio, Requests
- **Key Features / What I Built:**
    - Configured dynamic request throttling and rate-limiting algorithms conforming automatically to target robots.txt specifications.
    - Implemented a local HTML response caching system to minimize redundant external network requests during development.
    - Built strict Pydantic data normalization schemas to validate extracted DOM structures prior to database storage.
    - Integrated exponential backoff retry mechanisms to handle temporary HTTP 429 and 503 network drops gracefully.
- **Problem Solved / Impact:**
    - Achieved a 99.8% extraction success rate while completely avoiding IP rate-limits and target server disruption.
    - Reduced development re-fetch network overhead by 60% through local HTML document caching.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/polite-scraper](https://github.com/username/polite-scraper)

Sandboxed POSIX Shell Subsystem & Security Shield

- **1-Line Summary:** A high-performance, custom UNIX terminal subsystem engineered for system administrators requiring secure command routing and kernel-level sandboxing.
- **Tech Stack:** C++, POSIX System Calls, Custom Data Structures (Treap, Bloom Filter), Chroot Jail
- **Key Features / What I Built:**
    - Implemented a dual-hash DJB2/SDBM Bloom Filter malware shield for $O(1)$ pre-execution command threat detection.
    - Engineered command indexing and prefix-tree (Trie) auto-completion leveraging balanced Treap data structures.
    - Built a hardware-isolated chroot jail execution sandbox restricting filesystem access for untrusted binaries.
    - Designed custom memory-managed hashing algorithms to optimize system command lookup speeds.
- **Problem Solved / Impact:**
    - Eliminated unauthorized system command execution through real-time, memory-efficient Bloom Filter screening.

    - Reduced terminal command lookup and autocompletion latency to sub-millisecond speeds.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/secure-algorithmic-shell](https://github.com/username/secure-algorithmic-shell)

Enterprise AgriTech Relational Database System

- **1-Line Summary:** A multi-tenant AgriTech relational database platform engineered for dairy farm operators to centralize herd health, breeding cycles, and milk production.
- **Tech Stack:** T-SQL, Microsoft SQL Server, Database Design, Relational Modeling, React
- **Key Features / What I Built:**
    - Architected a normalized Relational Database Management System (RDBMS) featuring automated clinical event logging.
    - Engineered T-SQL stored procedures and triggers for real-time breeding optimization and lactation cycle tracking.
    - Implemented multi-table transaction rollback mechanisms ensuring zero data corruption across high-frequency inventory updates.
    - Integrated indexed views and query execution pipelines optimized for real-time farm management dashboards.
- **Problem Solved / Impact:**
    - Consolidated fragmented manual records into a centralized database capable of handling thousands of daily livestock data points.
    - Automated clinical treatment alerts, reducing missed veterinary follow-ups across herd populations.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/cattlefarm-dbms](https://github.com/username/cattlefarm-dbms)

CodeConnect: Visual x86 Assembly & System Integration Bridge

- **1-Line Summary:** An educational deep-tech platform designed for computer science students to visualize low-level x86 assembly logic and register operations via an interactive GUI.
- **Tech Stack:** C# WinForms, 32-bit x86 Assembly (NASM), P/Invoke (DllImport), C++
- **Key Features / What I Built:**
    - Architected an unmanaged-to-managed bridge using a C# P/Invoke layer (__cdecl/__stdcall) connecting WinForms to a custom 32-bit Assembly DLL.
    - Developed bitwise data packing algorithms in x86 Assembly manipulating upper/lower 4-bit register states for grid nodes.
    - Implemented double-buffered rendering in WinForms to deliver smooth, flicker-free low-level state visualizations.
    - Built custom stack frame management routines (push ebp/pusha) enforcing unmanaged memory safety.
- **Problem Solved / Impact:**
    - Replaced text-only assembly terminals with dynamic visual state tracking, clarifying low-level execution paths.

    - Resolved cross-architecture BadImageFormatException execution crashes by establishing strict x86 build target configurations.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/CodeConnect_COAL_Project](https://github.com/username/CodeConnect_COAL_Project)

Cryptographic Security & Steganography Engine

- **1-Line Summary:** An information security platform engineered for privacy-focused users to perform cryptographic key generation, file encryption, and hidden image steganography.
- **Tech Stack:** C#, .NET, Cryptography API, LSB Steganography, PBKDF2
- **Key Features / What I Built:**
    - Implemented PBKDF2 key derivation functions paired with AES encryption for key-based data protection.
    - Developed a Least Significant Bit (LSB) image steganography module to conceal encrypted payloads inside carrier images.
    - Built an algorithmic avalanche effect analyzer quantifying output bit-flips relative to single-bit key variations.
    - Integrated SHA-256 integrity headers to detect payload tampering during extraction.
- **Problem Solved / Impact:**
    - Provided end-to-end covert communication capability with cryptographic proof of message integrity and zero visual image degradation.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/secure-data-toolbox](https://github.com/username/secure-data-toolbox)

Containerized Microservice Task Management Engine

- **1-Line Summary:** A production-ready task orchestration API built for distributed teams to perform asynchronous CRUD operations with persistent storage.
- **Tech Stack:** Python, FastAPI, SQLModel, PostgreSQL 16, Docker, Docker Compose
- **Key Features / What I Built:**
    - Containerized a FastAPI microservice using multi-stage Docker builds to minimize deployment image size.
    - Configured Docker Compose orchestration managing isolated application and PostgreSQL database service networks.
    - Implemented persistent named volumes and environment-driven configuration management for zero-downtime restarts.
    - Designed relational SQLModel schemas with Pydantic serialization for strict API payload verification.
- **Problem Solved / Impact:**
    - Guaranteed 100% environment parity across local development and staging server deployments.
    - Reduced developer onboarding and environment setup time from hours to a single command.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/task-api-a3](https://github.com/username/task-api-a3)

Production RESTful Authentication API

- **1-Line Summary:** A secure identity management service built for web applications requiring robust JWT session verification and role-based access control.
- **Tech Stack:** Python, FastAPI, Supabase Auth, HTTPBearer, JWT, Pydantic
- **Key Features / What I Built:**
    - Engineered OAuth2/JWT verification middleware enforcing stateful and stateless session validation across REST endpoints.
    - Implemented HTTPBearer route-guarding logic to protect sensitive administrative actions from unauthorized access.
    - Built secure sign-up, credential authentication, and session revocation workflows connected to Supabase Auth.
    - Developed automated payload sanitization layers preventing credential injection attacks.
- **Problem Solved / Impact:**
    - Established a reusable, plug-and-play security architecture guaranteeing zero unauthorized endpoint access during system integration.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create /auth-api-w4](https://github.com/username/auth-api-w4)

Gamified Software AST Simulator & Engine

- **1-Line Summary:** A desktop programming simulator engineered for software learners to execute and debug logic puzzles within a controlled runtime loop.
- **Tech Stack:** C#, WPF, Entity Framework Core, Custom AST Parser, SQL Server
- **Key Features / What I Built:**
    - Built a custom Abstract Syntax Tree (AST) parser to validate, evaluate, and step through user-submitted code logic.
    - Configured Entity Framework Core state persistence to save real-time game loops and execution metrics.
    - Designed an event-driven WPF frontend utilizing responsive MVVM architecture for live simulation updates.
    - Integrated custom error handling layers trapping infinite loops and stack overflow conditions during parsing.
- **Problem Solved / Impact:**
    - Enabled real-time interactive code parsing and visual execution tracking with fault-tolerant sandbox protection.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create /code-to-survive-simulator](https://github.com/username/code-to-survive-simulator)

High-Performance Deterministic C++ AgriTech Engine

- **1-Line Summary:** A low-latency AgriTech management engine built for rural livestock facilities operating without continuous internet connectivity.
- **Tech Stack:** C++, Modern OOP, File Stream Serialization, Custom Caching Algorithms
- **Key Features / What I Built:**

    - Engineered a high-performance backend applying advanced Object-Oriented Programming (OOP) design patterns.
    - Developed a deterministic file-stream caching system ensuring instant local read/write operations without external database dependencies.
    - Built custom memory-efficient lookup trees to manage large-scale livestock inventory records in real time.
    - Implemented robust error recovery routines preventing data corruption during unexpected power losses.
- **Problem Solved / Impact:**
    - Delivered uninterrupted offline software capabilities for remote agricultural facilities requiring instant record access.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/agritech-cattle-farm-manager](https://github.com/username/agritech-cattle-farm-manager)

Minimalist Focus & Productivity Workspace

- **1-Line Summary:** A lightweight browser extension replacing the default new-tab page to eliminate digital distractions and optimize daily workflow focus.
- **Tech Stack:** JavaScript (ES6+), HTML5, CSS3, Web Storage API
- **Key Features / What I Built:**
    - Developed dynamic time-of-day greeting logic and real-time digital clock rendering algorithms.
    - Integrated persistent localStorage state tracking for custom daily goal setting.
    - Implemented a lightweight motivational quote engine cycling dynamic JSON data feeds.
    - Designed a dark-mode CSS UI optimized for zero render latency on tab creation.
- **Problem Solved / Impact:**
    - Reduced browser context switching and tab clutter, driving daily goal completion rates for productivity workflows.
- **Link:** GitHub Repo: [https://github.com/ahmadrayan-create/minimal-focus-dashboard](https://github.com/username/minimal-focus-dashboard)

Major Projects – Vibe-Coded Apps (AI-Orchestrated)

Philosophy: In 2026, using AI tools is not "cheating" — it is efficiency. The key is being able to explain every line of logic. Projects are labeled as *"Rapidly prototyped and deployed using AI-orchestration, focusing on architectural logic and prompt engineering to reduce development time by 70%."* This frames the developer as an efficient lead engineer rather than a passive user of AI.

Weather App (Android)

- **Tech Stack**: Android Studio, Cursor, AccuWeather API.
- Built a weather application with disaster-control features.
- Integrated real-time weather data and alerts for emergencies.
- Focused on architectural clarity and API orchestration for reliability.

Kisan Zarai Bazar – Agricultural Disease Detection App

- **Tech Stack**: Antigravity, Google AI Studio, Gemini API.
- Designed an application for farmers to upload crop disease images, analyze them through ai and then indicate exact disease, and then ai suggest the medicine for disease from brand inventory and its usage technique.
- Delivered solutions in both **Urdu and English**, with **voice output** for accessibility.
- Provided exact CTA product recommendations (e.g., pesticides, fertilizers).
- Increased sales while solving real farmer problems through AI-driven diagnostics.

Team Task Management Web Application

- **Platform**: Lovable.
- **Backend** : Firebase
- Developed a collaborative task management system for teams.
- Focused on workflow orchestration, user assignment, and progress tracking.

Other Vibe Coded Applications

- **pakistan-fuel-tracker [Private | TypeScript / React / Tailwind CSS]** 🔗 **Live App**: [https://pakistan-fuel-tracker-baoo.vercel.app/] [Live 🟢] — Real-time Pakistani fuel price tracker and economic intelligence terminal featuring live OGRA revision countdowns, commuter calculators, and logistics fleet cost projections. Engineered as a real-time, OGRA-integrated analytics engine to automatically calculate route fuel budgets and flag short-filling theft before transit departure. Delivers automated route cost budgeting via live tariffs, real-time litrage verification against exact route distances, and audit-ready expense reporting to eliminate manual spreadsheet errors and fleet profit leakage.
- **deepbuild-ai-services** [Private | TypeScript] — Full-stack AI services platform delivering autonomous web architectures, custom AI agents, and workflow automations to convert standard sites into revenue-generating assets.
- **eld-trip-planner** [Public | TypeScript / Django / React] 🔗 **Live App**: [ ] [Live 🟢] — Full-stack Electronic Logging Device (ELD) routing and compliance engine engineered to automate Hours of Service (HOS) calculations and map optimized commercial transit paths.
- **lawledge-platform-1** [Private | TypeScript / React] 🔗 **Live App**: [ ] [Live 🟢] — Civic engagement web portal unifying public legal resources, emergency readiness alerts, digital complaint filings, and real-time volunteer coordination.
- **BlueBarn** [Private | TypeScript / React] 🔗 **Live App**: [ ] [Live 🟢] — "The Google for Farms" Desi-Modern AgriTech platform tailored for Pakistani dairy operations, featuring automated milk management, animal digital twins, and financial analytics.

- **VectorGlide-3D** [Private | JavaScript / MERN / Three.js] 🔗 **Live App**: [ ] [Live 🟢] — High-fidelity 3D drone navigation simulator translating multivariable space curves into real-time Frenet-Serret frame calculations and visual kinematic vectors.
- **Deep-Build-App** [Public | TypeScript] — Central application repository hosting modular, ready-to-deploy AI applications, custom integrations, and client-facing AI automations.
- **CodeAlpha_VisionTracker-YOLOv8-ROI-Analytics** [Public | TypeScript / Python / YOLOv8] — Computer vision analytics system fusing YOLOv8 detection with Deep SORT tracking for trajectory visualization and algorithmic Region of Interest (ROI) monitoring.
- **CodeAlpha_IntentBot-Hierarchical-SemanticRetrieval** [Public | TypeScript / S-BERT] — Conversational FAQ agent built on S-BERT embeddings and hierarchical intent matching, utilizing Mahalanobis Distance for dynamic fallback retrieval.
- **CodeAlpha_AlphaTranslator-S2S-ContextualGlossary** [Public | TypeScript] — Low-latency bi-directional Speech-to-Speech (S2S) translation engine featuring a pre-API Contextual Glossary to guarantee 100% domain-specific vocabulary accuracy.
- **ahmadrayan-portfolio** [Public | TypeScript] — Interactive developer portfolio showcasing full-stack systems, architectural blueprints, and engineering projects.
- **FlyRank-Portfolio-Build** [Public | HTML / CSS] — Frontend web build delivering clean UI interfaces and assignment deliverables for industrial internship tasks.
- **task-api-a3** [Public | REST API] — Backend microservice built for structured task management, user execution pipelines, and automated endpoint testing.
- **ToDoApp** [Public | C#] — Desktop task management application implementing clean OOP architecture and local state persistence.

Other Projects (Courses, Personal Work, and Vibe Coding)

- **n8n Workflows**:
  - Built ~10 workflows integrating APIs, LLMs, and apps like Google Apps and Pexels.
  - Examples: automated task manager, day planner, note taker, and advanced workflow automations.
- **Portfolio Website with AI Agent**:
  - Created a complete portfolio site using vibe coding.
  - Integrated a trained chatbot AI agent as a **sales representative and customer support expert**. Using N8n Custom ai agents
  - Enabled live chat and meeting booking via n8n workflows and webhook connections. With calendar, etc.
  - Demonstrated mastery of AI agent training and workflow integration.
- **Make.com Workflows**:
  - Automated social media posting across WhatsApp, LinkedIn, Instagram using APIs.

    - Built a web scraping tool that finds supplier emails, writes outreach emails, sends them, and manages replies.
    - Leveraged tools like Appify for scraping and orchestration.
- **Data Analytics Capstone Project**:
    - Analyzed stock prices dataset (2000–present).
    - Thousands of rows of keggele dataset.
    - Cleaned, filtered, and worked on them in power BI.
    - Built dashboards, cleaned large datasets, and performed advanced analytics.
    - Demonstrated speed and efficiency in handling massive datasets, proving strong data analysis skills.

Important Subjects Studied – BS Computer Science (Air University, Multan)

- **Programming Fundamentals** – Core concepts of structured programming and problem-solving.
- **Discrete Mathematics** – Logic, sets, relations, functions, and graph theory.
- **Object-Oriented Programming (C++)** – Classes, inheritance, polymorphism, and modular design.
- **Digital Logic and Design** – Circuit design, Boolean algebra, and computer architecture simulation.
- **Visual Programming** – C#, Windows Forms, ASP.NET APIs, Razor Pages, XML integration.
- **Data Structures & Algorithms (DSA)** – Advanced C++ implementations, algorithmic efficiency, and complexity analysis.
- **Probability and Statistics** – Statistical modeling, hypothesis testing, and regression analysis.
- **Information Security** – Cryptography, hashing, steganography, and secure system design.
- **Full Stack Web Development** – MERN stack (MongoDB, Express.js, React.js, Node.js).
- **Computer Networks** – Protocols, communication models, and network security.
- **Computer Organization & Assembly Language** – Instruction sets, memory hierarchy, and low-level programming.
- **Multivariable Calculus** – Advanced mathematical foundations for computational modeling.
- **Software Engineering** – SDLC, design patterns, testing, and project management.

Seminars Attended

- Participated in multiple live seminars covering **business, management, computer science, artificial intelligence**, and related fields.

- Gained exposure to diverse perspectives and industry practices, enhancing both technical and professional knowledge.

Other Experiences and Skills

- **AI/ML Engineering Intern | SafeX Solutions** (July 23, 2026 – September 22, 2026) — Focused on translating machine learning theory into robust, production-ready ML pipelines designed to withstand real-world data friction and variance. Refined automated preprocessing routines and feature validation layers to ensure clean input data feeds, and implemented iterative model versions focusing on prediction accuracy and baseline performance metrics.
- **Backend AI Engineering Intern | FlyRank AI** (July 2026 – September 2026) — Focused on backend AI systems architecture—building reliable retrieval-backed flows, structured-output pipelines, and evaluation harnesses to ensure production-grade reliability. Defined robust API endpoints, strict payload schemas, standardized status codes, and error-handling layers, while establishing verifiable RAG audit trails between source data and AI responses.
- **Global Nobel Intern | Nobel Navigators** (June 2026 Cohort – Present) — Selected for the intensive, multi-stage global internship program focused on advanced professional leadership development, strategic problem-solving, and cross-cultural project execution in distributed remote environments.
- **AI Virtual Intern | CodeAlpha** (November 1, 2025 – November 30, 2025) — Successfully completed a Virtual Internship Program in Artificial Intelligence, recognized with a Letter of Recommendation by Founder & CEO Sorticri for exceptional analytical skills, rapid technical adoption, and high productivity.
- **Public Speaking & Oratory Excellence:** Exceptionally strong public speaking and debate capabilities honed through leadership roles as Team Coordinator in the university Debating Society, as well as serving as Chair of UNHRC and Academics Head at Air University's Model United Nations.
- **Institutional Coordination & Correspondence:** Proven administrative competency in managing official university communications, email flows, and cross-functional project groups in the capacity of Class Representative.
- **Commercial Engagement & Execution:** Demonstrated customer-facing communication and sales acumen through managing high-traction campus events, including leading a Summer Fest commercial stall that achieved a 50% profit margin.
- **Team Leadership & Mentorship:** Effective in stakeholder communication, technical team leadership, and collaborative problem-solving across academic cohorts and entrepreneurial ventures (DeepBuild and Blue Barn).

- **Adaptive Development**:
    - Demonstrated ability to learn new frameworks rapidly. For example, capable of building a complete Flutter app within a week using AI tools, even without prior experience.
- **AI Tools Expertise**:

    - Hands-on experience with a wide range of AI platforms for over a year, applied in real projects.
    - Tools include: Supabase, Firebase, Cursor, Lovable, Antigravity, Claude, Copilot, Gemini, Grok, Google AI Studio (apps, text-to-audio, video, etc.), Stitch, Opal AI, Vapi, Bolt, Emergent, Manus, Higsfield, Kling, Suno AI, N8N, Make.com, Replit, Perplexity, Kimi, Apify, Kaggle, and more.
    - Skilled in media generation, coding, workflow automation, and agentic AI orchestration.

Prompt Engineering & LLMs

- **Extensive Experience:** Over 100s of hours of hands-on prompting, tuning, and iterative model interaction with large language models since 2024.
- **Core Expertise:** Proficient in precision role definition, multi-modal media generation, advanced prompt engineering frameworks, and modern "vibe coding" execution paradigms.
- **Universal Adaptability:** Fully adaptive capability to orchestrate and leverage LLMs at their absolute peak performance and accuracy across any arbitrary technical, engineering, or business use case.

- **Financial Knowledge**:
    - Active in **forex and cryptocurrency trading** for 3 years.
    - Strong understanding of financial systems, market dynamics, and trading strategies.
- **Creative Media Skills**:
    - Video editing experience since 2019 using CapCut.
    - Skilled in video editing, picture grading, and creative content production.
- **AI Development Experience**:
    - Built multiple projects, websites, and applications using AI tools.
    - Practical application of AI in real-world contexts, demonstrating innovation and problem-solving ability.

Major Certifications and Courses

Professional Data Analytics Course – Leverify (Feb 2025)

- Completed a professional live course with daily assignments, evaluations, and expert instruction.
- Gained proficiency in **Microsoft Sheets, Google Sheets, Looker Studio, Power BI**, and data gathering/analysis workflows.
- Made multiple projects related to excel, looker studio, Power bi, data analyzing etc.
- Capstone Project: Analyzed the dataset of Italian stock prices (2000–present).

    - Built dashboards, cleaned large datasets, and performed advanced analytics.
    - Delivered actionable insights through fast, efficient data pipelines.
- Ranked among the **Top 10 scorers** in the course.

Vibe Coding and AI Agents Course – Leverify (July 2025)

- Selected through evaluation; only **40 participants out of 1000** were admitted.
- Pakistan's first-ever live course on **Vibe Coding and AI Agents**.
- Curriculum covered:
    - Fundamentals of AI, prompt engineering, and LLMs.
    - Building apps and websites using tools such as **Cursor, ChatGPT, Copilot, Grok, Lovable, Bolt, Firebase, Supabase, Firebase Studio**, and more.
- Key Projects:
    - **Weather App (Android)** and **Team Task Management Web Application**.
    - Built ~10 workflows in **n8n**, including task automation, day planner, note taker, and text-to-image generation.
    - Final project: Developed a complete **portfolio website** with an integrated **chatbot AI agent** acting as a sales representative and customer support expert, trained via n8n workflows and webhook connections.
- **Make.com Projects**:
    - Automated social media posting workflows across WhatsApp, LinkedIn, Instagram.
    - Built a web scraping tool that automatically finds supplier emails, writes outreach emails, sends them, and manages replies using **Appify**.
- Successfully completed the course and received **personal appreciation from the instructor** for outstanding performance.

Elite AI Cohort & Top Build Recognition – Leverify Quest *(August 2025)*

- **Selective Selection:** Selected out of **1,000+ applicants** into Pakistan's inaugural live AI Agents & Vibe Coding cohort (top 40 builders).
- **Top Build Award:** Honored with the **Top Build of the Cohort** award for the capstone project, which directly laid the operational and architectural foundation for **DeepBuild AI**.
- **Core Technical Execution:** Architected stateful autonomous agents, real-time webhooks, and multi-step business logic workflows via n8n and Make.com to automate complex operational bottlenecks.

Major Certifications and Courses – Outskill

Mastermind AI Generative AI Class – Outskill (Aug 2025)

Duration: 26+ hours, 3-day intensive workshop

Instructor: Vaibhav Sinistey (leading journalist in the AI industry), People from google, Microsoft, Netflix.

- Learned the **fundamentals of AI**, including chat models, prompt writing, prompt engineering, and chatbot usage.
- Gained hands-on exposure to multiple AI tools such as **VSPI, Suno AI**, and other automation platforms.
- Training included **AI automation workflows**, web browsing automation, and task orchestration.
- Learned from a **production house CEO** how to create visual content using AI — from prompts to full movies and advertisements. Using hives field etc.
- Built **custom AI agents** using Make and developed **custom GPTs** for specialized tasks.
- Covered the full spectrum of AI knowledge from **basics to advanced applications**, ensuring mastery of both theory and practice.

AI For Engineering – Outskill Professional Program (Oct 2025)

Objective: Designed to train participants as professional AI engineers.

Skills Taught:

- **Conversational AI**: Claude – reasoning and dialogue models.
- **Research AI**: Perplexity – information retrieval and synthesis.
- **Multimodal AI**: Gemini – Google's advanced multimodal framework.
- **Workflow Integration**: Google AI Studio – prompt design and orchestration.
- **Open Source Models**: Hugging Face – transformers and deployment.
- **Applied AI Development**: Genspark – real-world AI applications.
- **Productivity Integration**: Notion – AI-powered workspace.
- **Workflow Automation**: Make, Zapier – agent building and task automation.
- **Coding Assistance**: Copilot – AI-driven productivity in development.
- **Content Generation**: Manus – AI writing and editing.
- **Research Tools**: Notebook LM – summarization and literature review.
- **Data Analysis**: Statistical modeling and AI-driven insights.
- **Professional Applications**: Resume optimization, interview prep, research, marketing, product building, design, content writing.
- **Business & Strategy**:
    - Generating Organic Sales – AI-driven growth strategies.
    - LinkedIn Optimization – AI-assisted networking and branding.
    - Cold Emailing – AI-powered outreach.
    - Management Consulting – AI for decision support.
    - Meta Ads & Google Ads – AI-optimized campaign management.
    - Kairos MBA Masterclass – AI in business strategy and leadership.
    - Monetization Mastery 101 – AI in business intelligence, ad optimization, and operational efficiency.

Anthropic Certificates

- **AI Fluency: Framework & Foundations (Completed):** Collaborated with AI systems effectively, efficiently, ethically, and safely. Certificate uploaded and shared on LinkedIn.
- **Claude 101 & Claude Code 101 (In Progress / Pending):** Exploring everyday workflows, Claude Code integration, API development, and Model Context Protocol (MCP) server building.

Skills Learned from Online Courses and Applied Projects

Programming Languages

- **C++** – Advanced proficiency in low-level systems engineering, POSIX system calls, custom data structures (Bloom Filters, Treaps), and memory-managed algorithms.
- **C# / .NET** – Enterprise desktop development utilizing WinForms, WPF, MVVM architecture, Entity Framework Core, and unmanaged P/Invoke (DllImport) assembly integration.
- **TypeScript & JavaScript (ES6+)** – Modern web development for Next.js, React, and browser extensions using asynchronous APIs and Web Storage.
- **Python** – Backend microservice development with FastAPI, Pydantic data validation, Asyncio, BeautifulSoup4 web scraping, and LLM API integrations.
- **SQL (T-SQL & PostgreSQL)** – Relational database architecture, normalized schema design, stored procedures, multi-table transactions, and ACID-compliant event logging.
- **32-bit x86 Assembly (NASM)** – Low-level hardware interfacing, register-level bitwise manipulation, and stack frame management.
- **Java** – Object-oriented software design and core application architecture.
- **HTML5 & CSS3** – Responsive frontend interface design using modern layout engines and Tailwind CSS.

Modern Development Stacks

- **MERN Stack** – MongoDB, Express.js, React.js, Node.js for full-stack web development.
- Exposure to other modern frameworks and stacks for scalable application development.

AI & Automation Skills

- **Prompt Engineering** – Writing effective prompts, role definition, and chatbot usage.
- **Custom AI Agents** – Built agents using N8N, Make.com, and custom GPTs.

- **Workflow Automation** – Automated web browsing, data handling, content generation, and multi-step processes.
- **Gemini Gems Training** – Applied AI orchestration for diverse tasks:
    - **Productivity** – Automated workflows, reminders, and document generation.
    - **YouTube Automations** – Content workflows, insights extraction, video summaries.
    - **Custom CV Makers** – Recruiter-optimized resumes and portfolio entries.
    - **Tutoring Assistants** – AI tutors for coding, mathematics, and exam prep.
    - **Appointment Booking** – Automated scheduling systems.
    - **Web Scraping** – Data extraction using Apify.
    - **Lead Generation** – Automated prospecting, outreach, and engagement.
    - **Research Agents** – Structured insights from articles, reports, and datasets.
    - **Content Creation** – Blogs, social media posts, marketing copy.
    - **Design & Branding** – Portfolio graphics, ad creatives.
    - **Data Analysis & Reporting** – Dashboards, statistical summaries, BI reports.
    - **Custom AI Agents** – Specialized GPTs for financial tracking, project management, customer support.

Applied AI Deployment

- Implemented a **custom Gemini Gem** at a University Model United Nations (MUN) event to guide students acting as my replaceable as i was as in academics team so it guided students in place of me.
- Demonstrated applied AI deployment, user-centric design, and innovative automation in academic settings.

Professional Experience

Global Nobel Intern | Nobel Navigators ***(June 2026 – Nov 2026)***

- **Program Overview:** Selected for an intensive, multi-stage global leadership and professional development fellowship.
- **Collaboration & Execution:** Engaged in rigorous high-frequency collaborative modules spanning foundational global leadership, strategic problem-solving, and cross-cultural project execution in distributed remote teams.

Backend AI Engineering Intern | FlyRank AI *(July 2026 – Sept 2026)*

- **Role Focus:** Engineered production-grade backend AI systems architecture, building reliable retrieval-backed flows, structured-output pipelines, and evaluation harnesses to guarantee system reliability.

- **API Contracts & Design:** Defined robust FastAPI endpoints, strict payload schemas, standardized status codes, and error-handling layers prior to implementation.
- **Retrieval & Grounding (RAG):** Developed retrieval-backed answer flows and structured output pipelines (extraction, classification, drafting), establishing a verifiable audit trail between source data and AI responses.
- **Evaluation & Operations:** Implemented lightweight evaluation rubrics, custom test scripts, and observability plans to transition experimental AI features into predictable production services.
- **Resilience & Failure Analysis:** Integrated authentication wrappers and automated retry mechanisms under strict remote-first builder standards.

AI/ML Engineering Intern | SafeX Solutions *(July 23, 2026 – September 22, 2026)*

- **Role Focus:** Translated machine learning theory into robust, production-ready ML pipelines designed to withstand real-world data friction and variance.
- **Pipeline & Data Optimization:** Refined automated preprocessing routines and feature validation layers to ensure clean, high-throughput input data feeds for core models.
- **Model Architecture & Evaluation:** Implemented and tested iterative AI/ML model versions, focusing heavily on optimizing prediction accuracy, baseline performance metrics, and variance reduction.
- **Production-Minded Execution:** Bridged algorithmic development with structured system workflows, ensuring seamless scalability and applicability for distributed enterprise environments.

AI Virtual Intern | CodeAlpha *(November 1, 2025 – November 30, 2025)*

- **Recognition:** Awarded Letter of Recommendation by Founder & CEO Sorticri for exceptional analytical skills, rapid technical adoption, and high productivity in AI development tasks.

Venture & Technical Consultancy

Founder & Lead AI Solutions Architect | DeepBuild AI Services *(August 2025 – Present)*

- **Agency Overview:** An independent technical consultancy and AI engineering agency delivering custom web development, agentic workflow automations, and bespoke AI applications. *(Positioned as an active engineering exploration and high-impact freelance/consultancy practice).*
- **Agentic Workflow Automation:** Architected n8n and Make.com automation sequences integrating autonomous LLM reasoning engines with external APIs and Apify web scrapers.
- **Custom Web Development:** Engineered high-performance web applications utilizing Next.js, React, and Tailwind CSS, featuring streaming UIs for zero-latency user experiences.
- **Bespoke AI Applications:** Built specialized AI prototypes (RAG tutors, ReAct function-calling agents) and resolved architectural bottlenecks in client production applications.

- **Rapid Prototype & Delivery:** Utilized AI-assisted engineering methodologies (Cursor, v0) to compress full-stack development lifecycles from weeks to days while managing the entire agency deployment lifecycle independently on Vercel.

Startup

DeepBuild AI Services (Just recently started working on it from last year August -2025)

Role: Founder & Lead AI Solutions Architect

- **Live Agency URL:** https://deepbuild-ai-services.vercel.app/
- **1-Line Summary:** An AI engineering agency and strategic consultancy delivering custom web development, agentic workflow automations, bespoke AI application engineering, and technical advisory.
- **Tech Stack:** TypeScript, Next.js, React, Tailwind CSS, Vercel, n8n, Make.com, OpenAI API, Anthropic API, Apify, Python, Cursor, v0

Core Agency Services & Technical Prototypes

- **Agentic Workflow Automation:** Architected n8n and Make.com automation sequences connecting complex business processes with autonomous LLM reasoning, external APIs, and Apify web scrapers.
- **Custom Web Development:** Built high-performance, brand-tailored websites and web applications utilizing Next.js, React, and Tailwind CSS, featuring streaming UIs for zero-latency user experiences.
- **Bespoke AI Applications & Engineering:** Developed custom AI prototypes (RAG tutors, ReAct function-calling agents) and engineered solutions to troubleshoot, optimize, and resolve issues in existing AI applications.
- **AI Strategy & Consultancy:** Formulated technical evaluation frameworks to consult business owners on identifying high-ROI operational bottlenecks solvable via AI automation.

Engineering Benchmarks & Execution

- **Rapid Prototype Delivery:** Established an AI-assisted engineering methodology (Cursor/v0) to compress web and AI application build cycles from weeks to days.
- **Multi-Model Integration Architecture:** Benchmark-tested over 100 specialized AI models across unified interface frameworks to evaluate real-world latency, cost, and output accuracy for client use cases.

- **End-to-End Delivery:** Managed the entire agency lifecycle independently—from technical client discovery and system design to frontend development, API security, and Vercel cloud deployment.